

Technical Document #693: Data Engineering - Comprehensive Overview

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Data governance establishes policies and procedures for managing data assets. It covers data privacy, security, and compliance.

Data lakes store raw data in its native format. They support structured, semi-structured, and unstructured data at any scale.

Data warehouses store structured data optimized for analytical queries. They use columnar storage and support complex aggregations.

Schema evolution handles changes in data structure over time. Schema registries manage schema versions and ensure backward compatibility.

Data quality ensures data is accurate, complete, and consistent. Data validation, cleansing, and monitoring are essential components.

Data pipelines automate the movement and transformation of data between systems. Apache Airflow and Prefect are popular workflow orchestration tools.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Cybersecurity threats continue to evolve in complexity and scale. Organizations must invest in robust security measures to protect sensitive data.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Key Takeaways:

- Data governance establishes policies and procedures for managing data assets.
- Data quality ensures data is accurate, complete, and consistent.
- Data lineage tracks the origin and movement of data through systems.